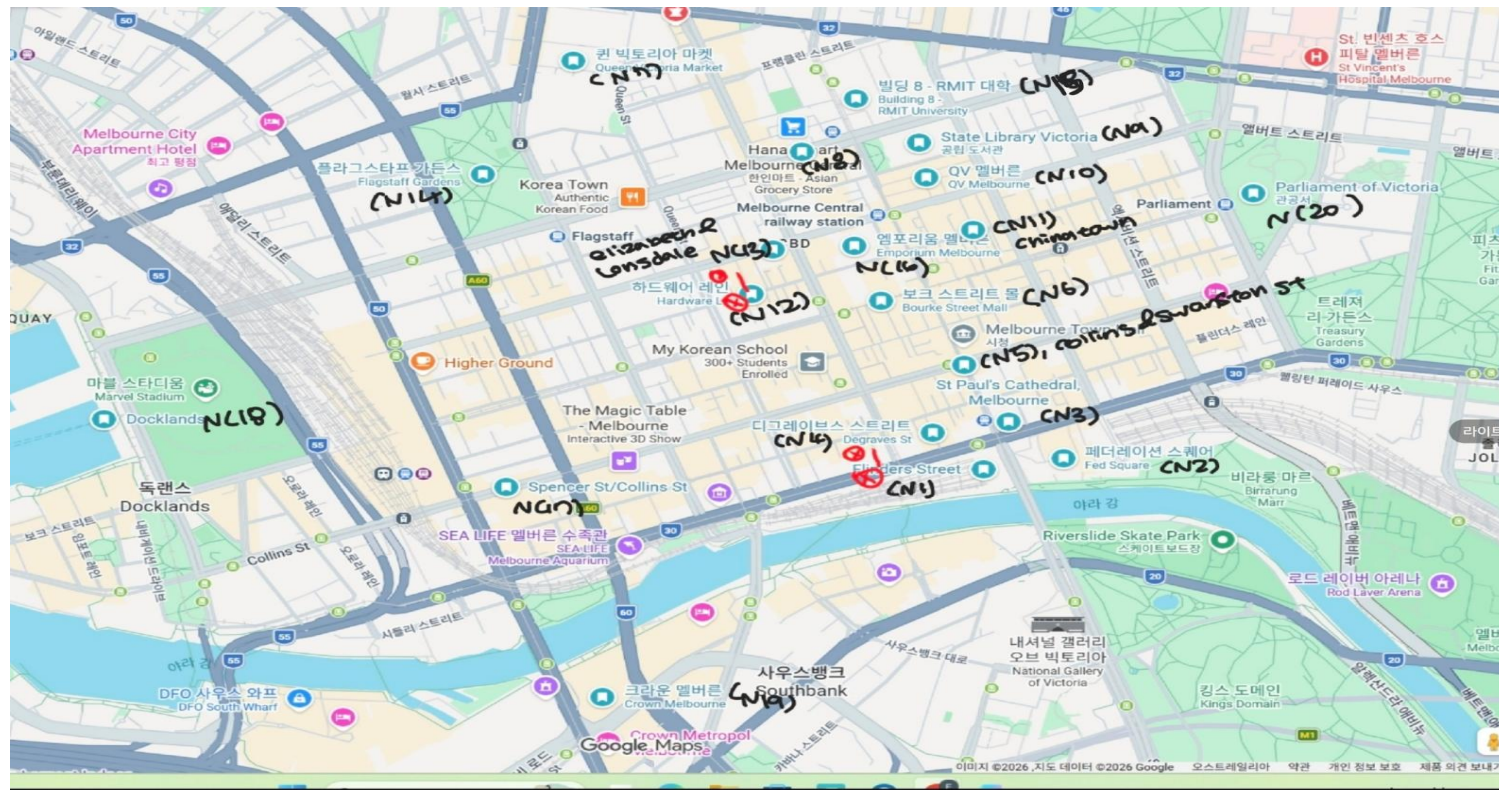




Landmark Inventory (20 Nodes)

ID	Landmark Name	Brief Description
N1	Flinders Street Station	[Start] Major transport hub with high pedestrian density.
N2	Federation Square	Open public space with complex movement patterns.
N3	St Paul's Cathedral	Historic site with restricted vehicle access for large agents.
N4	Degraves St Entrance	Narrow laneway; dynamic cost due to high cafe activity.
N5	Collins & Swanston Jct	Critical tram network hub with multiple intersection points.
N6	Bourke St Mall (East)	Pedestrianised zone with significant time-dependent congestion.
N7	Victoria Market	Major Grocery shopping center
N8	Melbourne Central Station	Major shopping and transport hub; a high-traffic node.
N9	State Library of Victoria	[Goal] Cultural precinct and the final delivery destination.
N10	QV Melbourne	Mixed-use development featuring complex internal path networks.
N11	Chinatown (Little Bourke)	High-demand dining district with narrow street constraints.
N12	Hardware Lane	Historic dining lane with evening access restrictions for vehicles.

ID	Landmark Name	Brief Description
N13	Elizabeth & Lonsdale Jct	Area with heavy bus traffic and potential congestion points.
N14	Flagstaff Gardens	Open Park area with potential surface and elevation constraints.
N15	RMIT Building 8	Educational precinct with high student pedestrian flow.
N16	Emporium Melbourne	Large-scale retail hub connecting multiple city blocks.
N17	Southern Cross Station	Regional transport hub marking the western boundary.
N18	Docklands Waterfront	Exposed waterfront area with environmental factors like wind.
N19	Crown Melbourne	Riverside pathway subject to occasional event-based closures.
N20	Parliament House	Government precinct requiring strict security-based rerouting.

Melbourne Landmark Graph (Edges)

The final graph contains 20 landmarks and 37 weighted path segments, which aligns with the recommended graph size for the assignment.

Access restrictions are present in certain areas such as Hardware Lane (N12) and Degrares Street (N4), where access for delivery trucks is limited due to pedestrian-only conditions. As a result, these locations are treated as constrained nodes in the graph.

The diagram represents an urban delivery network designed to analyse possible routes from Flinders Street Station (N1) to the State Library of Victoria (N9)

Costed Graph

Path costs were assigned to reflect travel difficulty in the abstract delivery environment. A base cost of 1 was used for standard road segments. A moderate cost of 2 was assigned to busy inner-city areas with likely congestion or complex movement patterns, such as Federation Square, Bourke Street Mall, QV, and Chinatown. A higher cost of 3 was assigned to constrained laneway segments, especially Degraves Street (N4) and Hardware Lane (N12), to reflect truck-specific access difficulty.

Cost 1 = Standard Road segment with relatively smooth truck movement

Cost 2 = Busy inner-city segment or major intersection with moderate congestion

Cost 3 = Constrained laneway segment with reduced suitability for truck access

Table: Melbourne CBD Delivery Network Adjacency List

Origin Node (ID)	Connected Successors (Cost)
N1 Flinders St Station	N2(2), N3(1), N19(1)
N2 Federation Square	N1(2), N3(2), N5(2)
N3 St Paul's Cathedral	N1(1), N2(2), N4(3), N5(1)
N4 Degraves St	N3(3), N6(3)
N5 Collins & Swanston	N2(2), N3(1), N6(2), N8(1), N10(1), N15(1)
N6 Bourke St Mall	N4(3), N5(2), N11(2), N12(3), N16(2)
N7 Victoria Market	N13(1), N14(1), N15(1)
N8 Melbourne Central	N5(1), N9(1), N10(2), N11(2), N13(1), N15(1), N16(1)
N9 State Library	N8(1), N10(2), N15(1)
N10 QV Melbourne	N5(1), N8(2), N9(2), N11(2), N15(1)
N11 Chinatown	N6(2), N8(2), N10(2), N12(3), N16(2)
N12 Hardware Lane	N6(3), N11(3), N13(3)
N13 Elizabeth & Lonsdale	N7(1), N8(1), N12(3), N14(1), N17(1)
N14 Flagstaff Gardens	N7(1), N13(1), N17(1)
N15 RMIT Building 8	N5(1), N7(1), N8(1), N9(1), N10(1)
N16 Emporium Melbourne	N6(2), N8(1), N11(2)

N17 Southern Cross	N13(1), N14(1), N18(1)
N18 Docklands	N17(1)
N19 Crown Melbourne	N1(1), N20(1)
N20 Parliament House	N9(1), N19(1)

Figure 1 shows the full weighted state-space graph of the Melbourne CBD delivery environment. Each node represents a landmark, and each weighted edge represents the travel cost of moving between two connected landmarks

Figure 1: Melbourne CBD Delivery Network - Full Weighted Graph

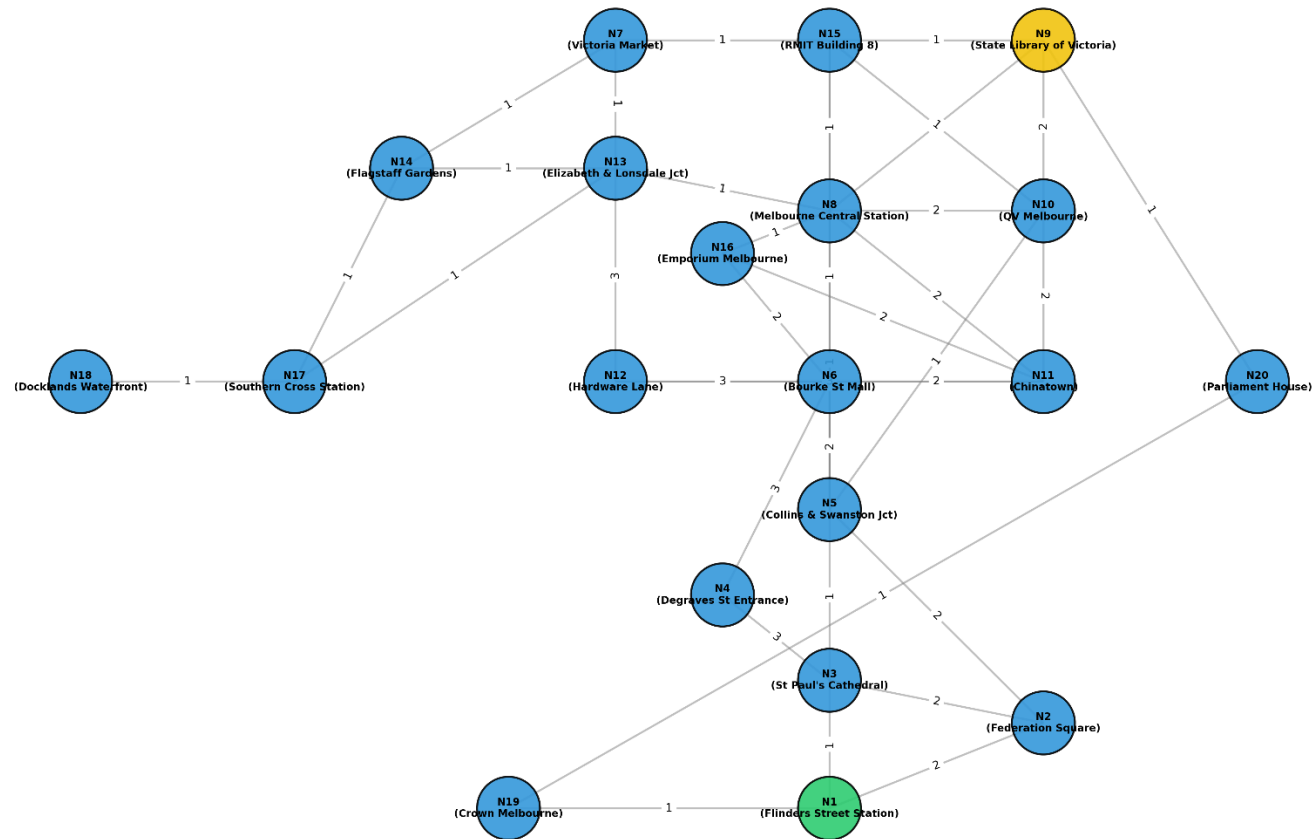


Figure 2: highlights the constrained nodes, Degrares Street (N4) and Hardware Lane (N12), which are less suitable for truck navigation due to narrow access and pedestrian-heavy conditions. These nodes remain feasible in the graph but are associated with higher traversal costs.

Graph with Constrained Nodes Highlighted (N4, N12)

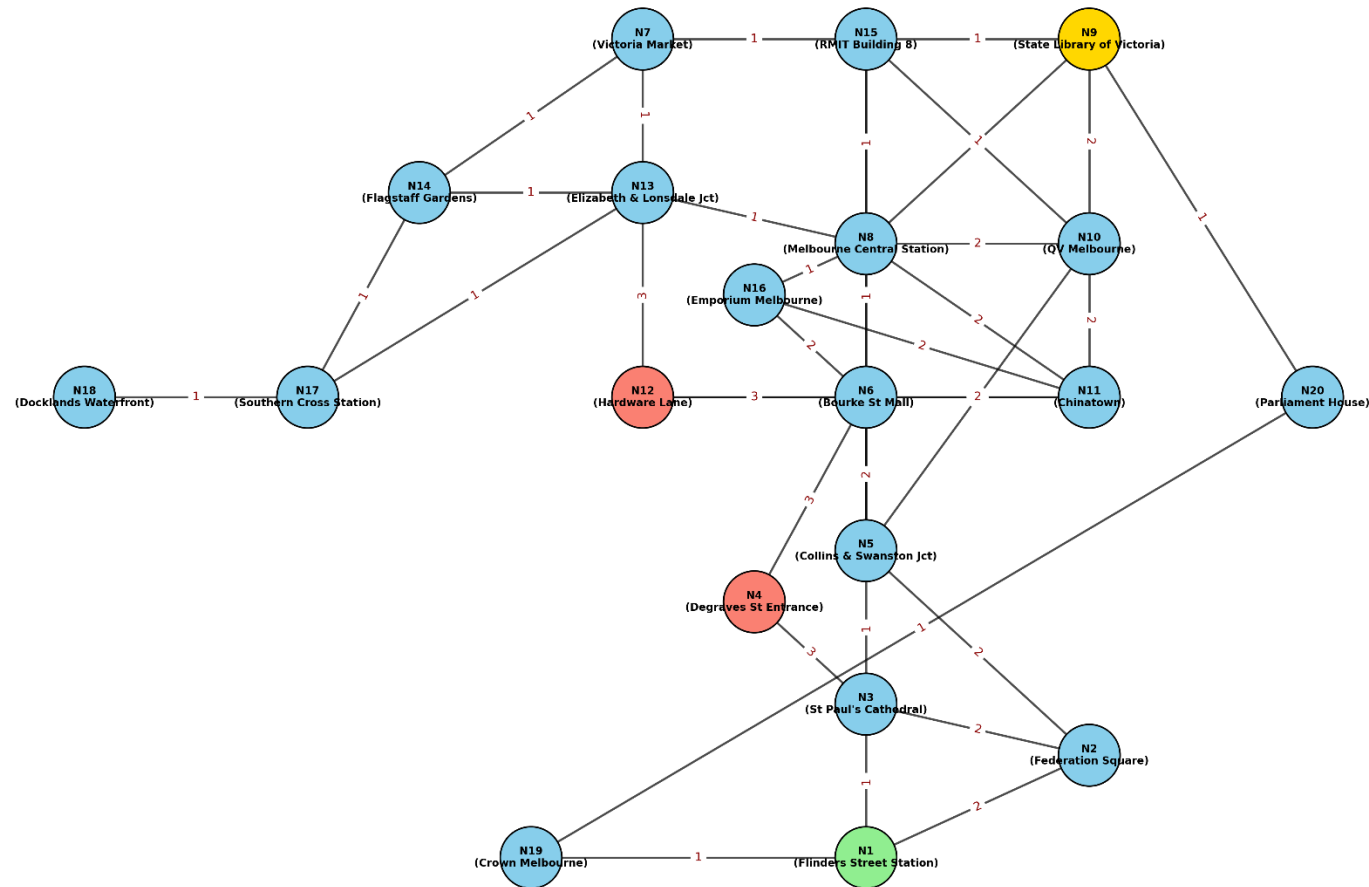


Figure 3 illustrates an example optimal route from N1 to N9 based on weighted path cost. The highlighted route demonstrates how the search process favours lower-cost paths while avoiding constrained areas where possible.

Figure 3: Optimal Route from N1 to N9 (Total Cost: 3)

summarised in Table 1.

SIT215 Assignment 1: Problem Formulation & Search Strategy

Problem Definition

The aim of this task is to design an autonomous delivery agent that can navigate through the Melbourne CBD and find an efficient path to its destination. The agent starts at Flinders Street Station (N1) and needs to reach the State Library of Victoria (N9).

The environment is a complex urban area with 20 landmarks, including transport hubs, shopping areas, and narrow laneways. Some areas experience heavy pedestrian traffic or temporary restrictions, which makes navigation more challenging.

The goal of the agent is to reach the destination efficiently while avoiding unnecessary delays.

PEAS Formulation

To design a rational agent, the PEAS framework is used to describe the task environment.

Performance Measure

The agent should minimise travel time, avoid highly congested areas such as Bourke Street Mall (N6), and successfully reach the goal node (N9) without failure.

Environment

The environment is the Melbourne CBD road network, including transport hubs, retail precincts, intersections, and constrained laneways. Some locations are more difficult for truck traversal due to congestion, narrow access, or heavy pedestrian activity.

Actuators

The agent uses movement controls such as moving forward, turning, and adjusting direction. It may also include communication systems to send updates.

In the abstract graph model, these actions are simplified as transitions from one connected node to another. Sensors

Sensors

The agent relies on GPS for location tracking, traffic information systems, and proximity sensors to detect nearby obstacles or pedestrians.

In the search model, sensing is abstracted as knowledge of the current node, connected successors, and associated edge costs.

Constraint Analysis

To better understand the environment, constraints can be divided into two categories: static and dynamic.

Static Constraints

These are time-invariant factors that define the permanent structure of the search space.

- **Structural Graph Connectivity:** The fundamental road network and predefined edges between the 20 landmarks.
- **Physical Dimension Restrictions:** As a heavy delivery truck, the agent must account for road width and turning radii. For instance, narrow laneways like **Degraves St (N4)** or **Hardware Lane (N12)** are modelled as static constraints where truck movement remains feasible but less desirable due to narrow road width, limited turning radius, and pedestrian-heavy conditions. These constrained segments are therefore assigned higher path costs rather than being completely removed from the graph.
- **One-Way Traffic Systems:** Fixed traffic flow directions in the Melbourne CBD that dictate the permitted actions at each node.

Dynamic Constraints

These represent environmental variables that change depending on the state and time, affecting the cost of the edges.

- **Pedestrian Surge & Congestion:** Areas such as **Bourke St Mall (N6)** and **Federation Square (N2)** exhibit high pedestrian density. These act as dynamic costs that can significantly slow down the agent or require temporary re-routing.
- **Event-Based Access Control:** Locations like **Crown Melbourne (N19)** or **Parliament House (N20)** are subject to sudden closures due to public events or security protocols, temporarily invalidating previously feasible paths.
- **Traffic Signal Cycles:** Stochastic delays at major junctions, such as **Collins & Swanston (N5)**, which influence the total travel time performance metric.

Note on Problem Relaxation: In the relaxed version of this problem used for search implementation, all **dynamic constraints** are removed to create a deterministic environment. Only the **static connectivity and physical access constraints** are enforced to ensure the benchmark algorithms can identify an optimal path within a stable state space.

Relaxed Problem Formulation

In order to apply search algorithms such as BFS and A*, the real-world delivery environment needs to be simplified into a more manageable form.

The Melbourne CBD is a complex and constantly changing environment, so directly applying search algorithms would be difficult. For this reason, the problem is relaxed by removing dynamic elements that are unpredictable in nature.

For example, factors such as pedestrian congestion at Bourke Street Mall (N6) or traffic delays at busy intersections like Collins & Swanston (N5) are not included in this model. These conditions change over time and would require a more advanced or probabilistic approach, which is beyond the scope of basic search algorithms.

Instead, the focus is placed on static constraints that remain consistent. This includes fixed road connections between landmarks and the general accessibility of locations for a delivery truck. In some cases, narrow or impractical paths, such as Degrares Street (N4), remain traversable in the relaxed model, but they are penalised through higher edge costs to reflect truck access difficulty.

By simplifying the environment in this way, the problem can be represented as a stable graph where each movement has a clear and consistent cost. This allows the search algorithms to operate effectively and makes it easier to evaluate their performance.

These modelling choices are important because they directly shape the search behaviour of the delivery agent. In this environment, path feasibility is not determined only by connectivity, but also by whether a route is practically suitable for a truck. By assigning higher costs to constrained laneways such as N4 and N12, the model captures the fact that some paths remain reachable but are less desirable. This abstraction allows the search algorithms to distinguish between merely possible routes and operationally sensible routes, which is essential for

intelligent behaviour in an urban delivery setting.

Overall, this relaxed model provides a clear starting point for analysing different search strategies before considering more complex real-world conditions.

State Space Modeling

The problem is represented as a graph where each landmark corresponds to a state.

- State: Each node (N1 to N20) represents a location in the city
- Initial State: Flinders Street Station (N1)
- Goal State: State Library of Victoria (N9)
- Actions: Moving from one node to another based on the defined edges
- Step Cost: Each movement has a weighted cost based on the corresponding edge in the graph. Standard road segments are assigned a cost of 1, busy inner-city segments are assigned a cost of 2, and constrained laneway segments are assigned a cost of 3.
- Transition Model: A move is allowed only if a weighted edge exists between the current node and the next node in the graph

This representation allows the problem to be solved using standard search algorithms.

Search Algorithm Analysis & Comparison

Two search strategies are used to evaluate the agent's performance: Breadth-First Search (BFS) and A* Search.

Breadth-First Search (BFS) was selected as the uninformed baseline because it provides a clear contrast with A*. BFS expands nodes level by level without using cost or heuristic information, which makes it useful for showing how an uninformed strategy behaves in the same

deterministic environment. This comparison is appropriate because it highlights the added intelligence of A*, particularly in a weighted urban graph where the number of steps and the total travel cost are not always the same.

New Heuristic Design

To improve A* performance in an urban delivery context, a customised heuristic called Truck-Optimized Priority Heuristic (TOP-H) is proposed.

Manhattan distance is used as a baseline because the Melbourne CBD can be approximated as a structured urban layout. It provides a simple estimate of the remaining distance, but it does not account for delivery-specific constraints.

The proposed heuristic is defined as:

$$h(n) = \text{Manhattan Distance}(n, \text{goal}) + \text{Traffic_Penalty}(n)$$

In this formulation, the Manhattan distance estimates how far a node is from the goal, while the Traffic_Penalty(n) reflects how suitable a location is for truck movement. Nodes on major roads are assigned lower penalties, whereas constrained areas such as narrow laneways and pedestrian zones are given higher penalties.

This encourages the agent to prioritise routes that are not only short in distance, but also operationally suitable.

Because the additional term represents a preference rather than a strict distance estimate, TOP-H is best interpreted as a guidance-oriented heuristic rather than a purely geometric lower bound. Therefore, Manhattan distance is treated as the baseline for theoretical comparison, while TOP-H is evaluated based on practical route quality.

This introduces a trade-off. While Manhattan distance provides a consistent estimate, it ignores real-world delivery constraints. In contrast, TOP-H incorporates domain-specific knowledge and guides the search away from less suitable areas. As a result, it improves decision making during search while still maintaining a reasonable and effective heuristic estimate.

Overall, TOP-H provides a more practical heuristic for this problem, as it aligns the search behaviour with real-world delivery requirements rather than relying only on geometric distance.

Implementation Overview

The environment is modelled as a weighted graph using the NetworkX library, where each landmark represents a node and each connection represents a possible route with an associated travel cost. The node data links each node ID to a real-world location, while the edge weights reflect delivery conditions. For example, constrained areas such as N4 and N12 are assigned higher costs to represent limited suitability for truck movement.

To evaluate routes, the implementation applies weighted shortest path logic, where the total path cost corresponds to $g(n)$ in search terminology. Although a built-in function is used, the behaviour aligns with cost-based search strategies, where lower-cost paths are prioritised over shorter but less suitable routes.

A drawing function is used to generate visual representations of the graph, including highlighting the start and goal nodes, constrained areas, and the final selected path.

Overall, the implementation reflects practical route selection by favouring operationally suitable paths rather than simply minimising the number of steps. This ensures that the agent avoids constrained areas when better alternatives are available.

A. Breadth-First Search (BFS)

Mechanism

BFS explores all nodes at the current level before moving to the next level. It expands nodes layer by layer.

Example Trace (Simplified)

Start at N1 → expand N2, N3, N19

Then expand N2 → N5 → N15 → N9

Path Found

N1 → N2 → N5 → N15 → N9

Evaluation

BFS is complete and guarantees the shortest path in terms of number of steps, but it does not consider edge costs and therefore may not produce the lowest-cost solution in a weighted graph. However, it requires a large amount of memory because it stores all nodes at each level.

B. A* Search

Mechanism

A* search uses both the actual cost ($g(n)$) and a heuristic estimate ($h(n)$) to guide the search.

In this problem, the heuristic is based on the proposed TOP-H function, which combines Manhattan distance and a traffic penalty.

Evaluation

A* is complete and can guarantee optimal solutions when used with an admissible heuristic. In this project, it is applied in a practical context where heuristic guidance is prioritised to improve route suitability. Compared to BFS, it explores fewer nodes by focusing on more promising paths. This makes A* more efficient and more suitable for real-world navigation scenarios.

BFS has a time and space complexity of $O(b^d)$, as it explores all nodes at each depth level without guidance. In contrast, A* reduces the effective branching factor through heuristic guidance, allowing it to reach the goal more efficiently while using less memory in practice.

Evaluation Table

Criterion	Breadth-First Search (BFS)	A* Search (Informed)
Completeness	Yes	Yes
Optimality	Not cost-optimal in a weighted graph	Yes (with admissible heuristic)
Search Efficiency	Expands all nodes level by level without direction	Guided towards goal using heuristic

Criterion	Breadth-First Search (BFS)	A* Search (Informed)
Nodes Expanded	High; explores many unnecessary paths	Lower; avoids irrelevant branches

Multiple Start–Goal Pair Evaluation

To evaluate the delivery model more comprehensively, three different start–goal pairs were tested on the same weighted graph. This allows the search behaviour to be examined across multiple delivery scenarios within the Melbourne CBD network.

Table 1. Comparison of weighted and BFS results

Start	Goal	Weighted Path Found	Total Cost	BFS Path	BFS Steps
N1	N9	N1 → N19 → N20 → N9	3	N1 → N19 → N20 → N9	3
N17	N10	N17 → N13 → N8 → N10	4	N17 → N13 → N8 → N10	3
N18	N15	N18 → N17 → N13 → N7 → N15	4	N18 → N17 → N13 → N7 → N15	4

The evaluation of the N17 to N10 route clearly demonstrates the fundamental difference between the two strategies. BFS identifies a path with only 3 steps (N17 → N13 → N8 → N10), as it only aims to minimize the number of nodes visited. However, A* accounts for the edge costs. Even if a path requires more steps, it selects the route with the minimum total cost. This confirms that while BFS is efficient for unweighted graphs, A* is a more suitable approach for urban delivery scenarios where traversal costs vary.

The results show that feasible delivery routes exist for all tested scenarios. However, the total path cost varies depending on the network structure, proximity to constrained nodes, and congestion-prone areas.

The BFS-style results demonstrate that while BFS can find paths efficiently in terms of steps, it does not consider edge weights and therefore does not always produce the minimum-cost solution in a weighted graph.

The consistency of feasible paths across multiple scenarios demonstrates that the model is robust and not limited to a single predefined route.

Conclusion & Recommendation

Based on the analysis, A* search is a more suitable solution for the Melbourne CBD delivery problem.

While BFS guarantees the shortest path in terms of the number of steps, it does not necessarily produce the minimum-cost solution in a weighted graph because it does not consider edge weights. In addition, BFS expands many unnecessary nodes, making it less practical for real-world navigation.

In contrast, A* demonstrates more intelligent behaviour by using heuristic information to guide the search. With the proposed TOP-H heuristic, the agent is able to consider both distance and road suitability, which is important for a delivery truck operating in a busy urban area. This allows the agent to avoid narrow laneways and high-risk pedestrian zones while still maintaining an optimal path.

The results also show that the model performs consistently across multiple start–goal scenarios, indicating that it is robust and not limited to a single predefined route.

Therefore, A* is recommended as the final approach, as it provides a strong balance between efficiency, optimality, and real-world applicability. This makes it a more practical solution for urban delivery systems where both cost and environmental constraints must be considered.

References

NetworkX Developers. (2026). *Shortest paths*

https://networkx.org/documentation/stable/reference/algorithms/shortest_paths.html

Russell, S., & Norvig, P. (2022)

<https://aima.cs.berkeley.edu/>

Google. (2026). *My Maps*

<https://www.google.com.au/maps>

A Formal Basis for the Heuristic Determination of Minimum Cost Paths" (1968)

<https://ieeexplore.ieee.org/document/4082128>